

Supplementary Information of
“Dynamic Transfer of Chiral Edge States in Topological Type-II Hyperbolic Lattices”

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Supplementary Note 1: Derivation of imaginary coupling coefficient in coupled outer and inner CESs

As illustrated in Supplementary Figure 1a, two bases are coupled with coefficients κ_{oi} and κ_{io} , this system can be described by the spatially coupled mode equation as:

$$-i \frac{d}{d\varphi} \begin{pmatrix} a_o \\ a_i \end{pmatrix} = \begin{pmatrix} \beta_o & \kappa_{oi} \\ \kappa_{io} & \beta_i \end{pmatrix} \begin{pmatrix} a_o \\ a_i \end{pmatrix} \quad (\text{S1})$$

where a_o and a_i represent the amplitude of the waves at outer and inner edges, respectively. Considering this system as an effective continuous model, as shown in Supplementary Figure 1b. Taking an arbitrary section in such a model, as illustrated by gray box, we apply the energy conservation principle:

$$-\frac{d}{dt} \int \varpi dA = \oint \vec{S} \cdot \vec{n} d\vec{L} \quad (\text{S2})$$

where ϖ is the energy density in the box, $\vec{S}$ is the power flow density and $\vec{n}$ is the unit vector normal the boundary. When we set the length of the box infinite small, Supplementary Equation (S2) reduces to:

$$\lim_{\Delta\varphi \rightarrow 0} \frac{1}{\Delta\varphi} \oint \vec{S} \cdot \vec{n} d\vec{L} = - \lim_{\Delta\varphi \rightarrow 0} \frac{1}{\Delta\varphi} \frac{d}{dt} \int \varpi dA = 0 \quad (\text{S3})$$

Besides, the left side can be divided into two parts:

$$\begin{aligned} \lim_{\Delta\varphi \rightarrow 0} \frac{1}{\Delta\varphi} \oint \vec{S} \cdot \vec{n} d\vec{L} &= \lim_{\Delta\varphi \rightarrow 0} \frac{1}{\Delta\varphi} [S_o(\varphi + \Delta\varphi) - S_o(\varphi)] - \lim_{\Delta\varphi \rightarrow 0} \frac{1}{\Delta\varphi} [S_i(\varphi + \Delta\varphi) - S_i(\varphi)] \\ &= \frac{dS_o}{d\varphi} - \frac{dS_i}{d\varphi} = \frac{d|a_o(\varphi)|^2}{d\varphi} - \frac{d|a_i(\varphi)|^2}{d\varphi} = 0 \end{aligned} \quad (\text{S4})$$

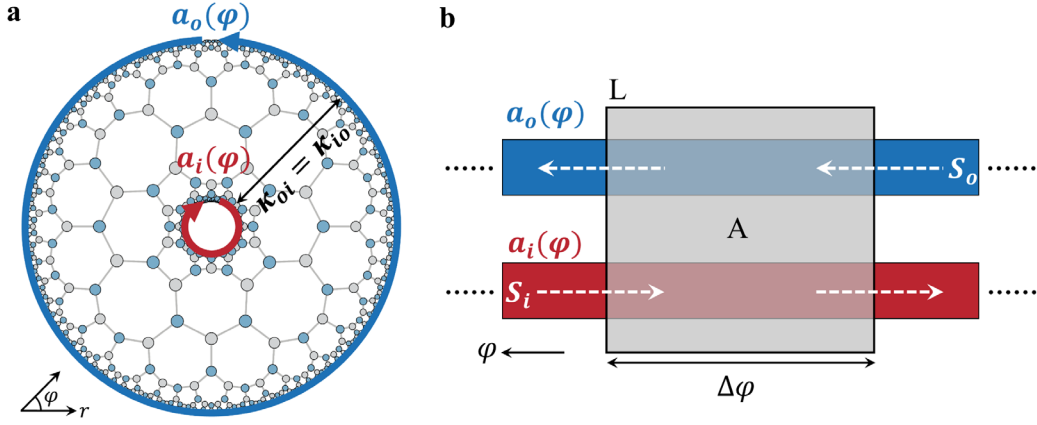
Substituting Supplementary Equation (S1) into Supplementary Equation (S4), we get:

$$\frac{d|a_o(\varphi)|^2}{d\varphi} - \frac{d|a_i(\varphi)|^2}{d\varphi} = i2\text{Im}[(\kappa_{oi} + \kappa_{io}^*)a_o^*(\varphi)a_i(\varphi)] = 0 \quad (\text{S5})$$

Thus we obtain:

$$\kappa_{oi} + \kappa_{io}^* = 0 \quad (\text{S6})$$

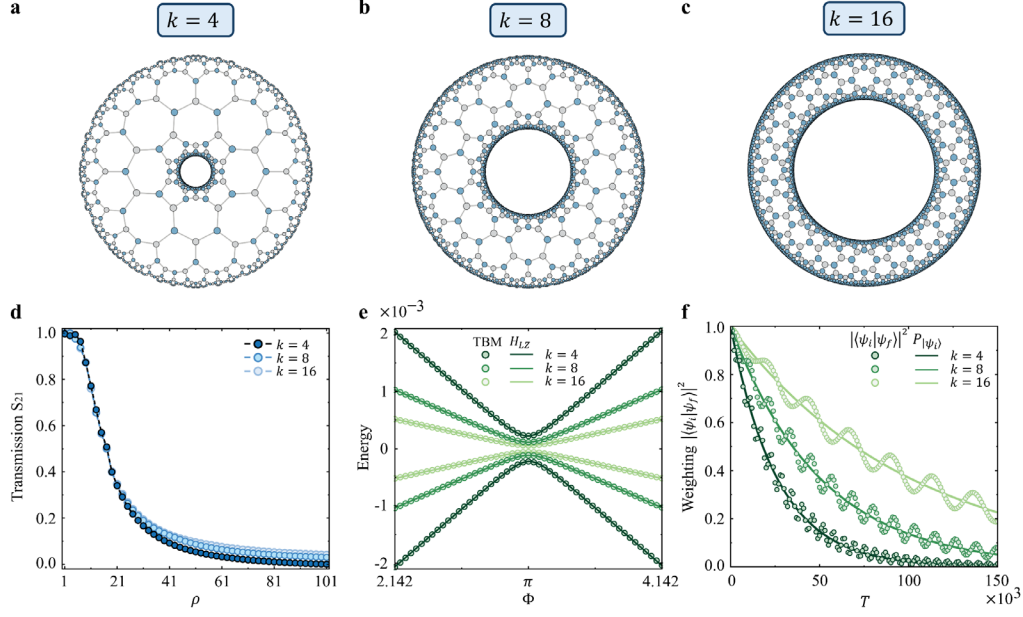
Given that the coupling process is reciprocal, i.e., $\kappa_{oi} = \kappa_{io}$, the coupling coefficients can be further derived as purely imaginary numbers as $\kappa_{oi} = \kappa_{io} = i\kappa$.



Supplementary Figure 1 | a Coupled outer and inner CESs in a type-II HCI. **b** Effective continuous model of **a**.

Supplementary Note 2: Comparison of efficiencies of dynamic transfer of outer and inner CESs via APT phase transition and LZ single-band pumping with different k

As shown in Supplementary Figure 2a-c, we select three distinct type-II HCIs with $k = 4, 8, 16$, respectively, and compare their efficiencies of dynamic transfer of outer and inner CESs via APT phase transition and LZ single-band pumping. As shown in Supplementary Figure 2d, with the same modulation strength ρ , there is only a minor difference in the transmissions S_{21} of outer-outer edge transport for three distinct modulated type-II HCIs. This indicates that the parameter k does not significantly affect the efficiency of the dynamic transfer of outer and inner CESs in the process of APT phase transition. In contrast, as shown in Supplementary Figure 2e, with the same modulation strength $\rho = 6$ and the same range of modulation phase Φ for three distinct modulated type-II HCIs, a larger k leads to a flatter band and a smaller gap size, which correspond to smaller fitted LZ parameters α and Γ . Therefore, according to the LZ formula $P_{|\psi_i\rangle} = e^{-\pi\Gamma^2/\eta}$, for the same number of evolution steps T , larger k leads to lower efficiency of LZ single-band pumping, as shown in Supplementary Figure 2f.



Supplementary Figure 2 | **a-c** Schematic illustrations of QWZ model on a $\{0.365, 8, 3\}$ ($k = 4$) (**a**), $\{0.604, 8, 3\}$ ($k = 8$) (**b**), and $\{0.777, 8, 3\}$ ($k = 16$) (**c**) type-II hyperbolic lattice, respectively. **d** Transmissions (S_{21}) of outer-outer edge transport as a function of modulation strength ρ for three different modulated type-II HCIs with $k = 4, 8, 16$, respectively. **e** Eigenenergy spectra as a function of the modulation phase Φ for three different modulated type-II HCIs with $k = 4, 8, 16$, respectively. The solid lines (dots) are analytic (numerical) results calculated from the LZ Hamiltonian H_{LZ} (TBM calculations). **f** The weightings of final state $|\psi_f\rangle$ on the base $|\psi_i\rangle$ as a function of number of evolution steps T . The solid lines (dots) are analytic (numerical) results calculated from the LZ formula $P_{|\psi_i\rangle}(|\psi_i|\psi_f|^2)$.

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